

§11.4 - The Comparison Tests

Lemma #1

Let (a_n) be a sequence of nonnegative numbers.

Then

1. the series $\sum a_n$ converges to a finite number,
2. or the series $\sum a_n$ diverge to infinity.

Theorem #1 (Basic Comparison Test)

Let (a_n) and (b_n) be sequences such that

$$0 < a_n \leq b_n$$

for all $n \in \mathbf{N}$.

Then

$$\sum a_n = \infty \quad \text{implies} \quad \sum b_n = \infty$$

and

$$\sum b_n \in \mathbf{R} \quad \text{implies} \quad \sum a_n \in \mathbf{R}.$$

As we argued with the Integral Test

$$S_n = \sum_{k=1}^n a_k$$

is increasing since $a_k > 0$.

If (S_n) is bounded then the partial sums converge by the MCT.

Otherwise,

$$S_n \rightarrow \infty.$$

If

$$0 < a_k \leq b_k \quad \ddagger \quad T_n = \sum_{k=1}^n b_k$$

then

$$0 < S_n \leq T_n.$$

If $T_n \rightarrow T$ then

$$S_n \leq T$$

and so (S_n) is bounded and converges.

Naturally,

$$S_n \rightarrow \infty \Rightarrow T_n \rightarrow \infty.$$

The MCT is an existence result. We do not know the value of

$$S = \sum_{k=1}^{\infty} a_k.$$

Remark

The condition

$$0 < a_n \leq b_n \text{ for all } n$$

in the Basic Comparison Test can be relaxed to hold for all n -large.

The condition

$$f \text{ decreasing on } [1, \infty)$$

in the Integral Test can be relaxed to f decreasing for all x -large.

Example #1

Determine if

$$\sum_{k=2}^{\infty} \frac{1}{\sqrt{k} - 1}$$

converges.

For all $k \in \mathbb{N}$ with $k > 2$

$$0 < \sqrt{k} - 1 < \sqrt{k}$$

$$0 < \frac{1}{\sqrt{k}} < \frac{1}{\sqrt{k} - 1}.$$

Choose

$$a_k = \frac{1}{\sqrt{k}} \quad \text{and} \quad b_k = \frac{1}{\sqrt{k} - 1}.$$

The p -series with $p = 1/2$ diverges to ∞

$$\sum_{k=2}^{\infty} \frac{1}{\sqrt{k}} = \infty \Rightarrow \sum_{k=2}^{\infty} \frac{1}{\sqrt{k} - 1} = \infty$$

by the Basic Comparison Test (BCT).

Example #2

Determine if

$$\sum_{k=2}^{\infty} \frac{\ln(k)}{k^2}$$

converges.

From the MVT,

$$0 < \ln(x) \leq x^{1/2} \text{ on } [4, \infty).$$

For $k > 3$

$$0 < \frac{\ln(k)}{k^2} \leq \frac{k^{1/2}}{k^2} = \frac{1}{k^{3/2}}$$

Choose

$$a_k = \frac{\ln(k)}{k^2} \quad \text{;} \quad b_k = \frac{1}{k^{3/2}},$$

then

$$\sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$$

is a convergent p -series since $p = 3/2 > 1$.

Thus

$$\sum_{k=2}^{\infty} \frac{\ln(k)}{k^2}$$

converges by the BCT.

Example #3

Determine if

$$\sum_{k=1}^{\infty} \frac{k}{k^2 + 2k + 3}$$

converges.

Here

$$0 < k^2 < k^2 + 2k + 3$$

and so

$$\frac{k}{k^2 + 2k + 3} < \frac{k}{k^2} = \frac{1}{k}.$$

Bounding our sum above by the divergent Harmonic series does not help.

Now

$$\frac{k}{k^2 + 2k + 3} \approx \frac{1}{k} \text{ for } k - \text{large}$$

suggests our sum diverges since it "looks like" the divergent Harmonic series.

We could argue

$$k^2 + 2k + 3 < 2k^2$$

to show

$$\frac{1}{2k} < \frac{k}{k^2 + 2k + 3}$$

for k -large but this takes work...

Theorem #2 (Limit Comparison Test)

Let (a_n) and (b_n) be sequences of positive numbers where

$$c = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}.$$

exists.

If $c > 0$ and finite then

$$\sum a_n \quad \text{and} \quad \sum b_n$$

either both converge or both diverge.

This is a special application of the Basic Comparison Test.

To use the test, you need a good guess as to what series to compare it to.

Idea: with $c > 0$

$$0 < \frac{a_k}{b_k} \approx c$$

$$\Rightarrow 0 < a_k \approx c \cdot b_k$$

$$\Rightarrow \sum a_k \approx \sum c \cdot b_k,$$

so the series are multiples of each other. They should both converge or diverge together.

Formally: With

$$\frac{a_k}{b_k} \rightarrow c > 0,$$

choose $N \in \mathbb{N}$ such that

$$\left| \frac{a_k}{b_k} - c \right| < \frac{c}{2}$$

for any $k > N$. Then

$$-\frac{c}{2} < \frac{a_k}{b_k} - c < \frac{c}{2} \Rightarrow \frac{c}{2} < \frac{a_k}{b_k} < \frac{3c}{2}$$

and so

$$0 < \frac{c}{2} b_k < a_k < \frac{3c}{2} b_k.$$

This means

$$\frac{c}{2} \sum_{k=N+1}^{\infty} b_k \leq \sum_{k=N+1}^{\infty} a_k \leq \frac{3c}{2} \sum_{k=N+1}^{\infty} b_k.$$

If $\sum a_k$ exists so must $\frac{c}{2} \sum b_k$ by the BCT.

This also means $\sum b_k$ exists.

If $\sum b_k$ exists so must $\frac{3c}{2} \sum b_k$, and then

$\sum a_k$ by the BCT.

Back to our example:

$$a_k = \frac{k}{k^2 + 2k + 3} \approx \frac{1}{k} = b_k$$

and

$$\begin{aligned} \frac{\frac{k}{k^2 + 2k + 3}}{\frac{1}{k}} &= \frac{k^2}{k^2 + 2k + 3} \\ &= \frac{1}{1 + \frac{2}{k} + \frac{3}{k^2}} \rightarrow 1 \end{aligned}$$

tells us $c = 1$. Hence

$$\sum_{k=1}^{\infty} \frac{k}{k^2 + 2k + 3} = \infty$$

by the Limit Comparison Test (LCT) since

$$\sum \frac{1}{k} = \infty.$$

Example #4

Determine if

$$\sum_{k=1}^{\infty} \frac{e^{2k} + k}{e^{5k} - k^2}$$

converges.

Notice

$$\begin{aligned}a_k &= \frac{e^{2k} + k}{e^{5k} - k^2} = \frac{e^{2k} \left(1 + \frac{k}{e^{2k}}\right)}{e^{5k} \left(1 - \frac{k^2}{e^{5k}}\right)} \\&= \frac{1}{e^{3k}} \cdot \frac{1 + \frac{k}{e^{2k}}}{1 - \frac{k^2}{e^{5k}}}\end{aligned}$$

and notice

$$\frac{k}{e^{2k}} \rightarrow 0 \quad \text{;} \quad \frac{k^2}{e^{5k}} \rightarrow 0$$

by L'Hospital. This yields

$$a_k \approx \frac{1}{e^{3k}} = b_k$$

since

$$\frac{a_k}{b_k} = \frac{\frac{e^{2k} + k}{e^{5k} - k^2}}{\frac{1}{e^{3k}}} \rightarrow 1.$$

Now $0 < 1/e^3 < 1$ and so

$$\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \left(\frac{1}{e^3}\right)^k$$

converges since we have a Geometric Series.

We have

$$\sum_{k=1}^{\infty} \frac{e^{2k} + k}{e^{5k} - k^2}$$

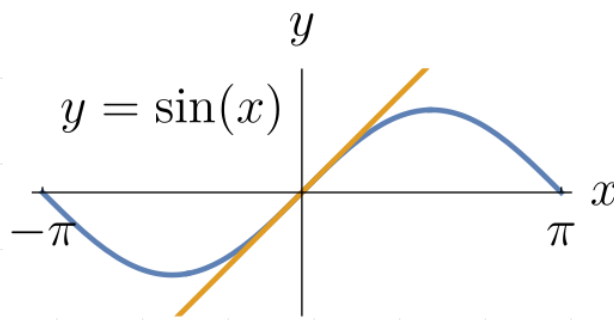
converges by the Limit Comparison Test.

Example #5

Determine if

$$\sum_{k=1}^{\infty} \frac{\sin(1/k)}{k}.$$

converges.



For $1/k$ "small",

$$\sin\left(\frac{1}{k}\right) \approx \frac{1}{k} \Rightarrow \frac{1}{k} \sin\left(\frac{1}{k}\right) \approx \frac{1}{k^2}$$

so let

$$a_k = \frac{1}{k} \sin\left(\frac{1}{k}\right) \quad \text{and} \quad b_k = \frac{1}{k^2}.$$

Now

$$\frac{a_k}{b_k} = \frac{\frac{1}{k} \sin\left(\frac{1}{k}\right)}{\frac{1}{k^2}} = \frac{\sin\left(\frac{1}{k}\right)}{\frac{1}{k}} \rightarrow 1$$

as $k \rightarrow \infty$ since

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

by L'Hospital's Rule or special trig limits in §3.3.

Then

$$\sum_{k=1}^{\infty} \frac{1}{k} \sin\left(\frac{1}{k}\right)$$

converges by the LCT with $\sum \frac{1}{k^2}$. A similar

LCT argument shows

$$\sum_{k=1}^{\infty} \sin\left(\frac{1}{k}\right) = \infty$$

since

$$\sum_{k=1}^{\infty} \frac{1}{k} = \infty.$$